

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_no_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_164430_e
- started_at_local: 2026-05-13 16:44:30
- finished_at_local: 2026-05-13 16:52:54
- duration_hhmm: 00:08
- duration_seconds: 504.172
- seed_offset: 4000
- replicate_label: e
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.164661	0.041579	0.10312	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.189222	0.008454	0.098838	0.862456	0.58	0.012982
atsp_failure_replay	failure	score_only	False	0.193715	0.009004	0.10136	0.846892	0.58	0.018339
atsp_failure_replay_compression	failure	novelty_gate	True	0.193715	0.009004	0.10136	0.856054	0.58	0.018143

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.164661, synthetic 0.041579, combined 0.10312.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.164661 across 4 instances; family means: ft=0.246054, ftv=0.272784, p=0.004804, ry=0.135002.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1690, "family": "ftv", "name": "ftv33", "optimality_gap": 0.314152}, {"best_known_cost": 1530, "cost": 1982, "family": "ftv", "name": "ftv38", "optimality_gap": 0.295425}, {"best_known_cost": 1473, "cost": 1840, "family": "ftv", "name": "ftv35", "optimality_gap": 0.249151}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.189222, synthetic 0.008454, combined 0.098838.
- Accepted-epoch count 4, mean accepted code novelty 0.862456, and final complexity 0.58.
- Adaptation efficiency 0.012982 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.189222 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.093884.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.193715, synthetic 0.009004, combined 0.10136.
- Accepted-epoch count 3, mean accepted code novelty 0.846892, and final complexity 0.58.
- Adaptation efficiency 0.018339 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193715 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193715, synthetic 0.009004, combined 0.10136.
- Accepted-epoch count 3, mean accepted code novelty 0.856054, and final complexity 0.58.

- Adaptation efficiency 0.018143 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193715 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

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Benchmark Suite Interpretation: Replay-aware ATSP

Summary of Conditions and Key Metrics

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Complexity	Accepted Epochs
atsp_no_replay	none	**0.1647*	0.0416	0.1031	0.0	0.78	1
atsp_random_replay	random	0.1892	**0.0085**	**0.0988**	**0.8625**	0.58	4
atsp_failure_replay	failure	0.1937	0.0090	0.1014	0.8469	0.58	3
atsp_failure_replay_compression	failure + compression aware	0.1937	0.0090	0.1014	0.8561	0.58	3

Key Observations

Transfer & Optimality Gaps

- **Best Transfer Performance:**

The atsp_random_replay condition shows the lowest mean transfer gap (0.0988) and best synthetic holdout gap (0.0085), indicating superior transfer compared to others.

- **TSPLIB Performance:**

The atsp_no_replay condition performs best on TSPLIB held-out instances with the lowest mean optimality gap of 0.1647, compared to higher gaps (~0.19) under all replay conditions.

- **Replay Impact on Transfer:**

Both random replay and failure replay conditions yield improved transfer gaps over no replay (0.0988 / 0.1014 vs. 0.1031), but at the cost of slightly increased TSPLIB gaps.

- **Compression:**

The compression-aware failure replay condition shows identical gaps to failure replay without compression, suggesting no degradation or improvement in transfer or optimality gaps due to compression pressure.

Code Novelty vs Transfer

- Replay conditions (random, failure, failure+compression) have **high code novelty** (mean ~0.85-0.86), but this does **not** correspond to better TSPLIB gaps; in fact, TSPLIB gaps worsen versus no replay (code novelty 0).
- The **no replay condition maintains zero code novelty** but has better TSPLIB transfer, indicating lexical/code novelty alone does not imply algorithmic invention or improved real-world transfer.

Complexity and Adaptation

- Replay conditions have **lower complexity** (0.58) compared to no replay (0.78), suggesting simpler final behaviors under replay.
- Adaptation efficiency is near zero for no replay vs small positive (~0.013-0.018) for replay, reflecting some efficiency gain with replay.

Condition Distinctions

- **No Replay (atsp_no_replay):**

Best TSPLIB transfer (optimality gap 0.1647) but worse synthetic and transfer gaps. Zero code novelty. Higher complexity.

- **Random Replay (atsp_random_replay):**

Best synthetic and transfer gaps (0.0085 and 0.0988). High code novelty (~0.86). Slightly worse TSPLIB gap (0.1892).

- **Failure Replay (atsp_failure_replay):**

Similar transfer gaps as random replay but slightly worse TSPLIB gap (0.1937). High code novelty (~0.85).

- **Failure Replay + Compression (atsp_failure_replay_compression):**

Same gaps as failure replay without compression. High code novelty. Compression pressure present with no observed gain.

Conservative Interpretation

- **Replay improves synthetic and transfer gaps**, indicating better adaptation and generalization on synthetic and transfer probes.
- **TSPLIB held-out gaps worsen with replay**, meaning transfer benefit is dataset dependent and replay may reduce performance on standard TSPLIB benchmarks.
- **Code novelty increases with replay but does not correlate with TSPLIB transfer gains**, implying code novelty signals code diversity rather than algorithmic improvement.
- **Compression-aware replay neither improves nor degrades transfer or optimality compared to failure replay alone**.

Recommendations

- Prioritize **no replay condition for TSPLIB benchmark scenarios** due to better held-out gaps.
- Employ **random replay for improved transfer and synthetic gap performance**, suitable for synthetic environment adaptation.
- Avoid overinterpreting code novelty as algorithmic novelty without corresponding gains in TSPLIB or synthetic gaps.
- Further investigation needed to understand why replay improves synthetic but not TSPLIB performance.

In summary:

While replay modes (random and failure) boost transfer and synthetic gap metrics and code novelty, the deterministic TSPLIB benchmark gaps indicate the no replay condition achieves better real-world transfer optimality. Therefore, evidence suggests replay methods help synthetic adaptation but can degrade general TSPLIB transfer performance; code novelty reflects diversification rather than guaranteed innovation. Compression-aware replay adds no significant benefit over standard failure replay.